

Sample works

February 29, 2024

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[2]: import os
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
from sklearn.model_selection import train_test_split
from skimage.transform import resize
from skimage import io
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

C:\Users\ask4s\AppData\Roaming\Python\Python39\site-packages\pandas\core\computation\expressions.py:21: UserWarning: Pandas requires version '2.8.4' or newer of 'numexpr' (version '2.8.1' currently installed).

```
from pandas.core.computation.check import NUMEXPR_INSTALLED
```

C:\Users\ask4s\AppData\Roaming\Python\Python39\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.4' currently installed).

```
from pandas.core import (
```

```
[5]: #Define constants
image_size = (150,150)
#Load image data and preprocess it
def load_images_from_folder(folder):
    images = []
    labels = []
    for subfolder in os.listdir(folder):
        subfolder_path = os.path.join(folder, subfolder)
        for filename in os.listdir(subfolder_path):
            img = io.imread(os.path.join(subfolder_path, filename))
            img = resize(img, image_size)
            images.append(img.flatten())
            labels.append(subfolder)
    return images, labels

#load data
train_data, train_labels = load_images_from_folder("F:\\PhD in Computer Science\\Old\\Datasets\\Alzheimer's Disease\\Alzheimer_s Dataset\\train")
```

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test_data, test_labels = load_images_from_folder("F:\\PhD in Computer_
↳Science\\Old\\Datasets\\Alzheimer's Disease\\Alzheimer_s Dataset\\test")
```

```
[10]: #Data Generation students learning data.
import numpy as np
import pandas as pd

# Set random seed for reproducibility
np.random.seed(42)

# Generate synthetic dataset
n_samples = 100
study_hours = np.random.uniform(0, 10, n_samples)
exam_scores = 50 + 10 * study_hours + np.random.normal(0, 5, n_samples)

# Create a DataFrame to store the dataset
df = pd.DataFrame({'Study Hours': study_hours, 'Exam Scores': exam_scores})

# Display the first few rows of the dataset
print(df.head())

# Save the dataset to a CSV file
df.to_csv('D:\\MMU\\2024\\Teaching\\Artificial_
↳Intelligence\\student_scores_dataset.csv', index=False)
```

	Study Hours	Exam Scores
0	3.745401	87.889247
1	9.507143	143.576394
2	7.319939	123.658198
3	5.986585	99.928004
4	1.560186	64.503505